MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (HONOURS) REGULAR EXAMINATION MARCH 2025
SECOND SEMESTER
Multi-Disciplinary Courses (MDC) - MG2MDCCFS101 - FUNDAMENTALS OF BLOCK CHAIN AND CRYPTOCURRENCY
(2024 ADMISSION ONWARDS)

Duration: 1.5 Hours Maximum Marks: 35

Part A
Short Answer Type Questions
Answer any 10 questions
Each question carries 1 mark

1. **Discuss the role of partition tolerance in a distributed system.**

ANSWER: Partition tolerance ensures that if a group of nodes is unable to communicate with other nodes due to network failures, the distributed system continues to operate correctly.

2. **Discuss about decentralized system.**

ANSWER: Decentralization is a core benefit and service provided by blockchain technology. By design, blockchain is a perfect vehicle for providing a platform that does not need any intermediaries and that can function with many different leaders chosen via consensus mechanisms.

3. **Discuss the role of consensus algorithms in blockchain.**

ANSWER: Consensus is a distributed computing concept that has been used in blockchain in order to provide a means of agreeing to a single version of the truth by all peers on the blockchain network.

4. **Explain the concept of decentralization in blockchain technology.**

ANSWER: This is a core concept and benefit of the blockchain. There is no need for a trusted third party or intermediary to validate transactions; instead, a consensus mechanism is used to agree on the validity of transactions.

5. **Define and examine asymmetric cryptography.**

ANSWER: Asymmetric cryptography refers to a type of cryptography where the key that is used to encrypt the data is different from the key that is used to decrypt the data. This is also known as public key cryptography. It uses both public and private keys to encrypt and decrypt data, respectively. Various asymmetric cryptography schemes are in use, including RSA, DSA, and ElGammal.

6. **Examine the purpose of Bitcoin Improvement Proposals (BIPs) in the blockchain ecosystem.**

ANSWER: These documents are used to propose or inform the Bitcoin community about the improvements suggested, the design issues, or information about some aspects of the bitcoin ecosystem. There are three types of Bitcoin improvement proposals.

7. **Discuss about block cipher.**

ANSWER: Block ciphers are encryption algorithms that break up the text to be encrypted (plaintext) into blocks of a fixed length and apply the encryption block-by-block. Block ciphers are generally built using a design strategy known as a Feistel cipher.

8. **Distinguish between Proof of Work (PoW) and Proof of Stake (PoS) consensus algorithms.**

ANSWER: Proof of Work (PoW): This type of consensus mechanism relies on proof that adequate computational resources have been spent before proposing a value for acceptance by the network. This scheme is used in Bitcoin, Litecoin, and other cryptocurrency blockchains.
Proof of Stake (PoS): This algorithm works on the idea that a node or user has an adequate stake in the system; that is, the user has invested enough in the system so that any malicious attempt by that user would outweigh the benefits of performing such an attack on the network.

9. **Select a reason why merchants should accept Bitcoin payments.**

ANSWER: Merchants should accept Bitcoin payments because it is becoming widely used, allows easy and quick mobile transactions through QR codes or links, is supported by most wallets, and enables international payments without needing banks.

10. **State the advantages of using a hardware wallet.**

ANSWER: Method used a tamper-resistant device to store keys. This tamper-resistant device can be custom built or with the advent of NFC-enabled phones, this can also be a Secure Element (SE) in NFC phones. Trezor and Ledger wallets (various types) are the most commonly used Bitcoin hardware wallets.

11. **Recommend a method to improve the scalability of the Bitcoin network.**

ANSWER: Bitcoin lightning network, is a solution for scalable off-chain instant payments.

12. **Rank Bitcoin payment methods based on transaction speed and cost.**

ANSWER: The fastest and cheapest Bitcoin payment method is the Lightning Network, offering near-instant transactions at minimal cost. On-chain transactions are slower and more expensive, especially during network congestion, while custodial solutions are fast but require trust in third parties.

Part B
Short Essay Type Questions
Answer any 3 questions
Each question carries 5 marks

13. **Compare the advantages and disadvantages of centralized and decentralized distributed systems?**

ANSWER: Centralized Distributed Systems

Advantages:

1. Simplified Control and Management: Centralized systems offer easier governance, as there is a single authority making decisions and enforcing rules.
2. Efficiency: Coordination is typically faster, with optimized resource usage and low overhead for consensus mechanisms.
3. Security Management: Easier to enforce security policies and updates due to the single point of control.

Disadvantages:

1. Single Point of Failure: If the central server fails, the whole system might be compromised or rendered inoperable.
2. Scalability Issues: Centralized systems can face bottlenecks as user demand grows.
3. Lack of Transparency: All data and control being held by a central authority can lead to trust issues.

Decentralized Distributed Systems

Advantages:

1. No Single Point of Failure: Multiple nodes provide redundancy and fault tolerance, enhancing reliability.
2. Increased Transparency and Trust: Especially in blockchain-based systems, every participant can verify the system's behavior independently.
3. Scalability Potential: With peer-to-peer architecture, decentralized systems can potentially scale more efficiently.

Disadvantages:

1. Complex Coordination: Achieving consensus across distributed nodes can be resource-intensive and slow.
2. Security Risks: Decentralized systems are susceptible to various attack vectors (e.g., Sybil attacks) without robust defences.
3. Maintenance Overhead: Harder to upgrade or enforce policies uniformly due to lack of central control.

14. **Explain the role of decentralization in blockchain technology and how it transforms traditional system.**

ANSWER: Decentralization is a fundamental principle of blockchain technology that significantly transforms traditional systems. In conventional centralized models, a single authority or intermediary controls data and operations, making systems vulnerable to failure, manipulation, and inefficiencies. Blockchain replaces this centralized structure with a decentralized network where control is distributed among multiple independent nodes. These nodes participate in decision-making through consensus mechanisms like Proof of Work (PoW), eliminating the need for trusted third parties. This shift enhances transparency, security, and fault tolerance by ensuring that no single point of failure can disrupt the system. Moreover, decentralization empowers users by giving them direct control over their data and transactions. It enables the development of new applications and paradigms where services can operate autonomously, transparently, and without centralized oversight. As a result, blockchain decentralization redefines how information systems are structured, promoting fairness, resilience, and trust across various sectors (also include methods of decentralization).

15. **Prove why Bitcoin is considered a decentralized currency and how it differs from central bank digital currencies (CBDCs).**
    Bitcoin is considered a decentralized currency due to its foundational architecture and principles, which are fundamentally different from central bank digital currencies (CBDCs). Here's a breakdown of why Bitcoin is decentralized and how it differs from CBDCs:
    Why Bitcoin Is Considered a Decentralized Currency
1. **No Central Authority:**
    - Bitcoin operates on a **peer-to-peer network** where no single entity, government, or organization controls the issuance, validation, or management of the currency.
    - Transactions are **verified by miners** and recorded on a **distributed public ledger (blockchain)**.
2. **Open Source Protocol:**
    - The Bitcoin protocol is open source, meaning anyone can **review, use, or contribute** to its development.
    - Changes to the protocol require **network-wide consensus**, not unilateral decisions.
3. **Distributed Ledger Technology (Blockchain):**
    - The blockchain is maintained by a **global network of nodes** that independently validate and store copies of all transactions.
    - This ensures **immutability** and **resilience** against tampering or central control.
4. **Consensus Mechanism – Proof of Work (PoW):**
    - Bitcoin uses a **PoW consensus mechanism**, requiring miners to solve complex computational puzzles to add blocks to the chain.
    - This secures the network without needing a trusted third party.
5. **Limited Supply:**
    - Bitcoin has a **fixed supply cap of 21 million coins**, and its issuance is governed by an algorithm—not human policy—ensuring **predictability** and **scarcity**.

16. **Analyse Confidentiality, Integrity, and Authentication in cryptographic systems.**

ANSWER: Cryptography is the science of making information secure in the presence of adversaries. It does so under the assumption that limitless resources are available to adversaries. Ciphers are algorithms used to encrypt or decrypt data, so that if intercepted by an adversary, the data is meaningless to them without decryption, which requires a secret key. Cryptography is primarily used to provide a confidentiality service.
In addition to a confidentiality service, cryptography also provides other security services such as integrity, authentication (entity authentication and data origin authentication),

and non-repudiation. Additionally, accountability is also provided, which is a requirement in many security systems.

Confidentiality

Confidentiality is the assurance that information is only available to authorized entities.

Integrity

Integrity is the assurance that information is modifiable only by authorized entities.

Authentication

Authentication provides assurance about the identity of an entity or the validity of a message. There are two types of authentication mechanisms, namely entity authentication and data origin authentication.

1. **Entity authentication**

Entity authentication is the assurance that an entity is currently involved and active in a communication session.

2. **Data origin authentication**

Also known as message authentication, data origin authentication is an assurance that the source of the information is indeed verified. Data origin authentication guarantees data integrity because if a source is corroborated, then the data must not have been altered.

Part C
Essay Type Questions
Answer any 1 question
Each question carries 10 marks

17. **Differentiate the different layers of a decentralized ecosystem and explain their functions.**

ANSWER: To achieve complete decentralization, it is necessary that the environment around the blockchain also be decentralized. The blockchain is a distributed ledger that runs on top of conventional systems. These elements include storage, communication, and computation. There are other factors, such as identity and wealth, which are traditionally based on centralized paradigms, and there's a need to decentralize these aspects as well in order to achieve a sufficiently decentralized ecosystem.

Storage

Data can be stored directly in a blockchain, and with this fact it achieves decentralization. However, a significant disadvantage of this approach is that a blockchain is not suitable for storing large amounts of data by design. It can store simple transactions and some arbitrary data, but it is certainly not suitable for storing images or large blobs of data, as is the case with traditional database systems. A better alternative for storing data is to use Distributed Hash Tables (DHTs). DHTs were used initially in peer-to-peer file sharing software, such as BitTorrent, Napster, Kazaa, and Gnutella.

Two primary requirements here are high availability and link stability, which means that data should be available when required and network links also should always be accessible.

The incentive mechanism for storing data is based on a protocol known as Filecoin, which pays incentives to nodes that store data using the Bitswap mechanism.

Communication

The internet (the communication layer in blockchain) is considered to be decentralized.

This model is based on unconditional trust of a central authority (the service provider) where users are not in control of their data. Even user passwords are stored on trusted third-party systems. Thus, there is a need to provide control to individual users in such a way that access to their data is guaranteed and is not dependent on a single third party. Access to the internet (the communication

layer) is based on Internet Service Providers (ISPs) who act as a central hub for internet users.
An alternative is to use mesh networks. Even though they are limited in functionality when compared to the internet, they still provide a decentralized alternative where nodes can talk directly to each other without a central hub such as an ISP.
Computing power and decentralization
Decentralization of computing or processing power is achieved by a blockchain technology such as Ethereum, where smart contracts with embedded business logic can run on the blockchain network. Other blockchain technologies also provide similar processing-layer platforms, where business logic can run over the network in a decentralized manner.

18. Recommend a secure setup process for installing and using a Bitcoin wallet.

ANSWER: 1. Choose the Right Wallet Type (Define wallets and its types in detail)

- Hardware Wallets (e.g., Ledger, Trezor): Best for long-term storage; they store keys offline, making them resistant to malware.
- Software Wallets:
  - Desktop Wallets: Offer better security than mobile/web wallets but are more vulnerable than hardware wallets.
  - Mobile Wallets: Convenient but less secure; best for small amounts.
  - Web Wallets: Most convenient but least secure; avoid for large holdings.
- Paper Wallets: Involve printing your keys and storing them physically; high risk if not handled correctly.

2. Secure Installation

- Verify Wallet Software: Download only from official websites or trusted sources. Check digital signatures or hashes to ensure the software hasn’t been tampered with.
- Use a Clean Computer: Install and set up your wallet on a device free from malware or spyware. Consider using a live operating system like Tails for maximum isolation.